Gene Essentiality from the Cell Graph

The backbone against flux balance analysis and Flux Cone Learning,

and what routing the prediction through the metabolic module does

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September 16, 2026

Abstract

Gene essentiality is the one endpoint on which a flux model has a published, non-trivial score: on the released Merzbacher 2025 yeast split, flux balance analysis reads an AUROC of 0.69 over all 1,121 labeled genes and Flux Cone Learning 0.73 on the 223 held-out genes. This document scores the cell graph transformer on the same held-out genes, puts FBA on every medium the ontology defines, and asks two questions of the metabolic module: whether a prediction read off its growth column ranks essential genes better than the backbone alone, and where the gradient goes when the module is one of two paths rather than the only one. Every number here is at a validation-selected epoch on a pinned test set, with a gene bootstrap and the across-seed range, and every table is generated from the result files.

Section status: ✓ ready ■ author-reviewed, keep stable ✗ not done, agents may edit freely

Contents

1 The endpoint, its labels and its splits ✗ . . . **2**
1.1 Vocabulary ✗ . . . 2
1.2 Where the labels come from ✗ . . . 2
1.3 The protocol inherited from 027, and what it got wrong ✗ . . . 2
2 The comparators, computed rather than quoted ✗ . . . **2**
2.1 Flux balance analysis on four media ✗ . . . 2
2.2 Flux Cone Learning on the same genes ✗ . . . 3
2.3 The classical ladder on the same split ✗ . . . 3
3 The backbone: splits, labels, weight decay and the shape of the curve ✗ . . . **4**
3.1 The curve was noisy because the validation set was small ✗ . . . 4
3.2 The split file costs training genes, and the cells are under-trained ✗ . . . 4
3.3 Weight decay and graph regularization do nothing here ✗ . . . 4
3.4 More labels from the genome make the held-out ranking worse ✗ . . . 4
3.5 Gradient norms ✗ . . . 6
4 The metabolic module: three ways to read essentiality off it ✗ . . . **6**
4.1 Round 2: the module on the stable backbone, with and without the penalties ✗ . . . 6
5 Open ✗ . . . **9**

1 The endpoint, its labels and its splits ✗

1.1 Vocabulary ✗

AUROC Area under the receiver operating characteristic, computed as the Mann-Whitney rank statistic with mid-ranks on ties: the probability that a randomly drawn essential gene scores above a randomly drawn nonessential one.

backbone The cell graph transformer over the nine-graph perturbed gene graph with `prot_T5_all` node features, read at the pooled perturbed tokens. The *pooled* arm is this backbone with a scalar head and no metabolic module.

gene bootstrap Resampling the scored genes with replacement (2,000 draws) and reporting the 2.5 and 97.5 percentiles of the statistic.

held-out genes The 223 genes Merzbacher 2025 held out of its own training (`test = 1` in the released split), of which 195 resolve as single-deletion genotypes in the build. Every model number in this document is on those 195 unless the gene set is named.

metabolic module The enzyme-constrained thermodynamic flux layer over yeast-GEM 9.0.2 that maps the perturbed representation to a flux vector; here it is read at the objective column `r_2111` (growth).

released labels The 1,121 essentiality calls in the released split. The column named `essential` is a viability flag (1.0 marks a nonessential gene); it is inverted once and the four class counts are asserted against the paper's Methods.

selected epoch The argmax of a centered five-epoch moving average of the validation AUROC. The held-out genes are scored once, at that epoch.

1.2 Where the labels come from ✗

The build is a union of the SGD essential-gene list (1,140 genotypes, all essential) and the single-mutant fitness screens (Costanzo 2016, Kuzmin 2018; 4,540 genotypes), 5,656 single-deletion genotypes in all. No fitness value enters any objective: the head is one logit scored by binary cross entropy against a 0/1 label, and every genotype whose gene carries no label arrives masked. Two label sources are used and named throughout:

Table 1: The split files, one per label source: genes per split with the essential count in parentheses. The test block is the released held-out genes under either source; validation is a class-stratified fifth of the remaining labeled genes at seed 0. Under the genome source the released label wins on every disagreement (12 genes).

split file	labels	train (ess)	val (ess)	test (ess)
`ess_fcl_val0.2_s0`	released (1,121)	629 (100)	158 (25)	195 (31)
`ess_genome_val0.2_s0`	released + SGD essential + viable fitness (5,795)	4,368 (879)	1,093 (220)	195 (31)

1.3 The protocol inherited from 027, and what it got wrong ✗

The wave-9 protocol pinned the held-out genes and let the data module draw train and validation at 80/10/10 over the remaining 5,461 records. Of those, 926 carry a label, so the validation set held 75 labeled genes with about 12 essential, on which a single AUROC read has a Hanley-McNeil standard error near 0.07; 10 percent of the labeled genes sat in a random test block nothing reports; and 4,535 unlabeled records were forwarded every epoch under a False mask. A split file replaces the draw: it names every record's split, is class-stratified, carries the labels it was stratified on, and the trainer drops every record it does not name.

2 The comparators, computed rather than quoted ✗

2.1 Flux balance analysis on four media ✗

The published FBA comparator is cobrapy `single_gene_deletion` with glucose as the carbon source, scored on all 1,121 labeled genes. The phenotype it is compared against was called on rich medium, and the model here is scored on the 195 held-out genes, so the comparison as published is neither on the same medium nor on the same genes. Table 2 recomputes it on yeast-GEM 9.0.2 under the exchange bounds the model ships with and under the three media the ontology defines, with the essentiality score being the negative knockout-to-wild-type growth ratio, rounded to 10^{-3} so

that the roughly 900 deletions that grow at 0.99 to 1.00 of the wild type rank as the tie they are (without the rounding two identical runs differed by 0.03 AUROC on the held-out genes).

Table 2: FBA single-gene deletion on yeast-GEM 9.0.2 (cobrapy 0.30.0), scored as the negative knockout-to-wild-type growth ratio rounded to 10^{-3}. AUROC on all 1,121 labeled genes, on the 223 held-out genes, and on the 195 held-out genes the model scores with a 2,000-draw gene bootstrap interval. 1121 of the labeled genes are in the GEM; the rest keep wild-type growth.

medium	WT growth (1/h)	lethal	all 1,121	held-out 223	our 195 [95% CI]
shipped bounds (glucose minimal, aerobic)	0.086	186	0.707	0.633	0.654 [0.56, 0.75]
SM, synthetic minimal	0.314	160	0.707	0.642	0.666 [0.57, 0.76]
SC, synthetic complete	0.543	91	0.719	0.653	0.666 [0.57, 0.76]
YPD approximation (no peptone)	0.535	104	0.715	0.653	0.664 [0.57, 0.76]

The medium moves the number of lethal deletions by a factor of two and the AUROC by about 0.01. The published 0.69 on all labeled genes is reproduced to within 0.02.

2.2 Flux Cone Learning on the same genes ✗

Recomputed from the authors' released per-gene predictions: 0.7308 on the 223 held-out genes (published 0.73) and 0.742 on the 195 genes the model scores. On those same 195 genes the released FBA predictions score 0.640, so the published pair overstates FBA by about 0.06 relative to a head-to-head on one gene set. The paper prints a third number, 0.78 ± 0.065 in the legend of its Figure 3a, which is the mean over five cross-validation folds drawn inside its 898 training genes; a model scored on the held-out genes is compared with the 0.73 line, never with that one.

2.3 The classical ladder on the same split ✗

The fitness and expression lines each carry a ladder of models that need no graph and no training beyond a solve, and essentiality gets the classification form of it: FBA's growth ratio with no fitting, then L_2 logistic regression, cosine nearest neighbors and a random forest on each of four external gene embeddings, and logistic regression on an embedding with the four FBA growth ratios appended. Every fitted rung is fit on the split file's training genes, its one hyperparameter chosen on the validation genes, and scored once on the 195 held-out genes, the protocol of every transformer cell in Sections 3 and 4 (Table 3).

Table 3: The classical ladder on the released-label split (`ess_fcl_val0.2_s0.json.gz`): fit on the train genes, hyperparameter chosen on the validation genes, scored once on the 195 held-out genes (31 essential) with a 2,000-draw gene bootstrap. B1 is FBA's growth ratio with no fitting.

rung	embedding	chosen	val AUROC	held-out AUROC [95% CI]
B0 prior	–	–	0.500	0.500 [0.50, 0.50]
B1 FBA growth	–	medium=shipped	0.683	0.654 [0.56, 0.75]
B1 FBA growth	–	medium=SM	0.710	0.666 [0.57, 0.76]
B1 FBA growth	–	medium=SC	0.726	0.666 [0.57, 0.76]
B1 FBA growth	–	medium=YPD	0.714	0.664 [0.57, 0.76]
B2 logistic	prot_T5_all	C=0.001	0.722	0.763 [0.67, 0.85]
B3 cosine kNN	prot_T5_all	k=25	0.772	0.684 [0.56, 0.81]
B4 random forest	prot_T5_all	max_depth=4	0.743	0.768 [0.68, 0.86]
B5 logistic + FBA	prot_T5_all	C=0.1	0.833	0.803 [0.72, 0.87]
B2 logistic	calm	C=0.1	0.790	0.737 [0.64, 0.82]
B3 cosine kNN	calm	k=10	0.731	0.692 [0.58, 0.80]
B4 random forest	calm	max_depth=8	0.785	0.715 [0.61, 0.81]
B5 logistic + FBA	calm	C=10	0.839	0.803 [0.72, 0.88]
B2 logistic	species_lm_5p_3p	C=0.001	0.631	0.571 [0.47, 0.68]
B3 cosine kNN	species_lm_5p_3p	k=50	0.631	0.588 [0.48, 0.69]
B4 random forest	species_lm_5p_3p	max_depth=8	0.689	0.603 [0.50, 0.71]
B5 logistic + FBA	species_lm_5p_3p	C=0.001	0.747	0.686 [0.58, 0.79]
B2 logistic	normalized_chrom_pathways	C=10	0.722	0.775 [0.68, 0.86]
B3 cosine kNN	normalized_chrom_pathways	k=50	0.687	0.584 [0.47, 0.70]
B4 random forest	normalized_chrom_pathways	max_depth=None	0.801	0.724 [0.62, 0.82]
B5 logistic + FBA	normalized_chrom_pathways	C=10	0.820	0.796 [0.70, 0.88]

Two rows set the bar for everything that follows. Logistic regression on the ProtT5 protein embedding alone scores 0.763, level with Flux Cone Learning's 0.742 and with the best round-1 transformer cell, and the same regression with the four FBA growth ratios appended scores 0.803 [0.72, 0.88]; the CaLM embedding with FBA reads 0.803 as well. Neither the embedding nor the solver reaches 0.80 alone. FBA's ratio is a feature the ontology computes for every gene

at no training cost, so the transformer is measured against 0.80 from here on, not against the published 0.73, and any cell that scores below the logistic rung is scoring below a model with 1,025 parameters.

3 The backbone: splits, labels, weight decay and the shape of the curve ✗

Every cell in this section is the pooled backbone at width 12 with one transformer block and a one-layer preprocessor (16,874 parameters), trained for 80 epochs at three initialization seeds with the split pinned. Round 1 (jobs 2149 to 2181) moves one lever at a time away from the cell that trains on the released-label split file (`ess_fcl_val0.2_s0`, 629 training genes, 158 validation genes): the old 75-gene validation draw, the genome-wide labels, weight decay at 0, 10^{-2} and 10^{-1}, batch 32, learning rate at 3×10^{-4} and 3×10^{-3}, the class weight 772/126, and graph regularization off. Table 4 carries every number; Figure 1 the curves.

Table 4: Round 1, the backbone. Width 12, one block, one-layer preprocessor, 80 epochs, seeds 2700 to 2702. Held-out AUROC on the 195 genes: the seed mean with its standard deviation, the three seeds, the gene bootstrap of the pooled per-gene logits, the epochs selected on validation, and the median absolute epoch-to-epoch step of the validation curve.

arm	mean	sd	seeds	pooled [95% CI]	selected epochs	val step
wave-11 protocol (75-gene val)	0.759	0.020	0.781, 0.755, 0.741	0.760 [0.67, 0.84]	18, 38, 14	0.0056
released labels, 158-gene val	0.725	0.019	0.745, 0.722, 0.708	0.738 [0.65, 0.83]	66, 77, 77	0.0026
weight decay 0	0.725	0.019	0.745, 0.720, 0.709	0.735 [0.64, 0.82]	66, 77, 77	0.0027
weight decay 1e-2	0.726	0.017	0.745, 0.722, 0.711	0.735 [0.64, 0.83]	65, 77, 77	0.0029
weight decay 1e-1	0.725	0.018	0.746, 0.721, 0.709	0.735 [0.64, 0.82]	66, 77, 77	0.0025
batch 32	0.765	0.005	0.766, 0.770, 0.759	0.764 [0.67, 0.85]	74, 73, 77	0.0032
genome labels (5,795 genes)	0.704	0.024	0.706, 0.726, 0.679	0.705 [0.61, 0.80]	77, 77, 69	0.0020
fcl_lr3e-3	0.686	0.043	0.718, 0.704, 0.637	0.713 [0.62, 0.81]	77, 77, 50	0.0024
fcl_lr3e-4	0.658	0.093	0.737, 0.682, 0.556	0.732 [0.64, 0.82]	77, 77, 74	0.0023
fcl_noreg	0.725	0.019	0.745, 0.722, 0.709	0.738 [0.65, 0.83]	66, 77, 77	0.0026
fcl_posw	0.759	0.016	0.764, 0.772, 0.741	0.769 [0.68, 0.86]	77, 75, 53	0.0034

3.1 The curve was noisy because the validation set was small ✗

The median epoch-to-epoch step of the validation AUROC is 0.0056 on the 75-gene draw, 0.0026 on the 158-gene split and 0.0020 on the 1,093-gene genome-wide split (Figure 1b). Two of the three width-24 cells of the previous protocol selected epochs 4 and 5 on validation peaks of 0.82 and 0.86 and scored 0.74 on the held-out genes; no cell on a split file selects before epoch 50. The old protocol's curve reads near 0.8 on its own 75 genes and the split cells read 0.65 to 0.72 on theirs, so the levels are not comparable, only the steps are.

3.2 The split file costs training genes, and the cells are under-trained ✗

The split-file cell scores 0.725 against the old protocol's 0.759 on the same held-out genes. It trains on 629 labeled genes where the old draw trained on roughly 740, and every one of its seeds selected epoch 66 to 77 of 80 with the validation curve still rising (Figure 1a). At batch 128 an epoch is five optimizer steps, so 80 epochs is 400 updates. The two levers that move all three seeds in the same direction are both more optimization: batch 32, four times the steps per epoch, lifts the seeds by 0.020, 0.048 and 0.051 and collapses their spread to 0.005; the class weight, which steepens the curve, lifts them by 0.019, 0.051 and 0.033. Round 3 combines the two and raises the epoch budget.

3.3 Weight decay and graph regularization do nothing here ✗

Weight decay at 0, 10^{-2} and 10^{-1} reproduces the base cell to the third decimal on every seed, and so does removing graph regularization. At 400 updates with learning rate 10^{-3}, a decay of 10^{-1} shrinks a weight by four percent; the graph term's λ is 6.5×10^{-5}. Neither is a lever on this endpoint at this budget. The learning rate at 3×10^{-4} and 3×10^{-3} is worse than 10^{-3} on the seed mean, with seed spreads of 0.09 and 0.04 that put the pooled numbers inside the base interval.

3.4 More labels from the genome make the held-out ranking worse ✗

The genome-wide file adds SGD's essential list and the viable single-mutant fitness records as nonessential, 4,368 training genes against 629, and scores 0.704 against 0.725, three of three seeds down. Its validation AUROC on its own 1,093

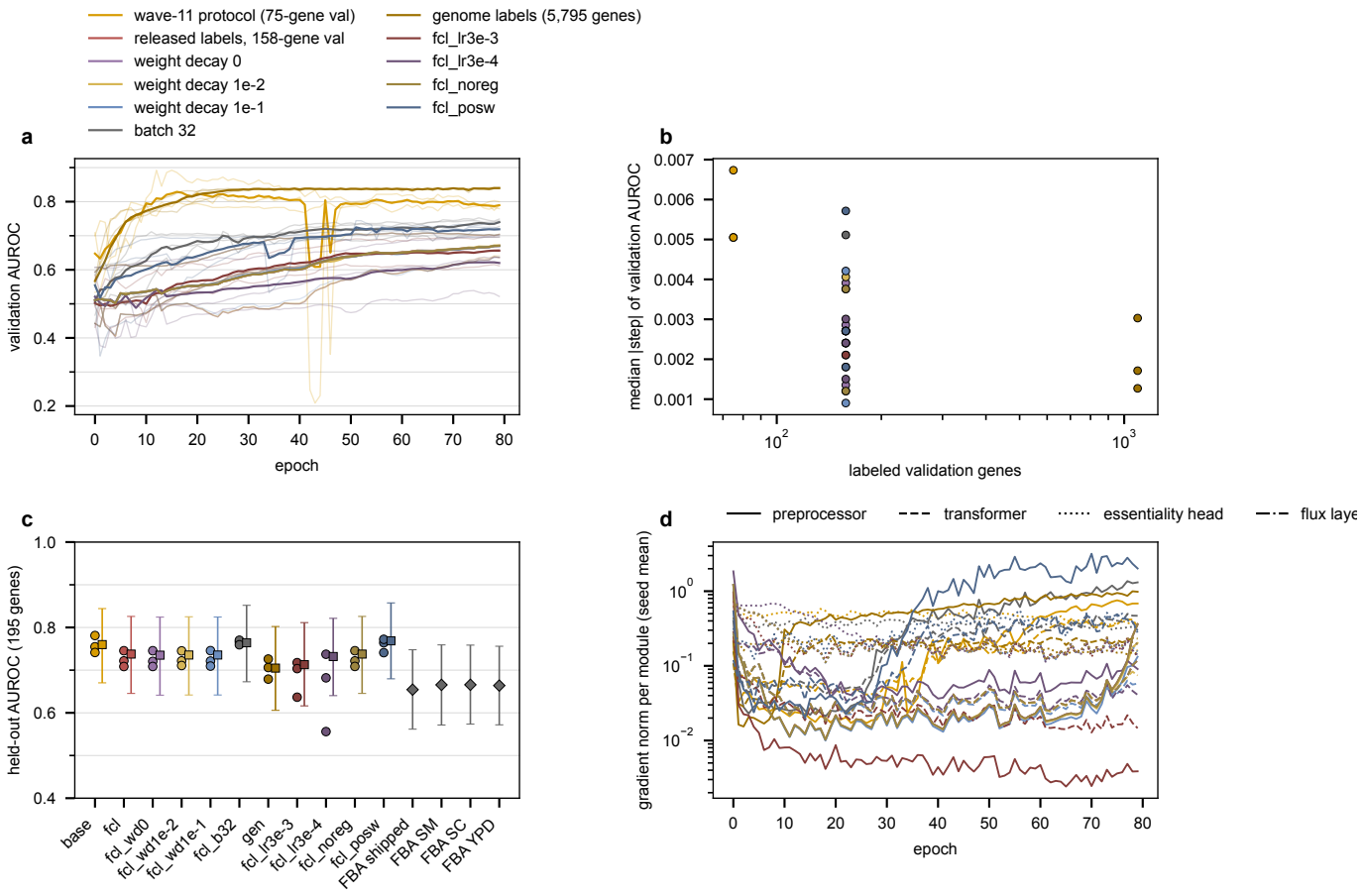


Figure 1: Round 1, the backbone. **a**, Validation AUROC per epoch, thin lines per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step of that curve against the number of labeled validation genes it was read on. **c**, Held-out AUROC on the 195 genes per cell, the seed mean, and the gene bootstrap of the pooled per-gene logits, beside FBA on each medium. **d**, Per-module gradient norm per epoch, seed mean, on the arms that logged it. Generated by `scripts/wave12_readout.py -round w12`.

genes is 0.84 while the released held-out genes read 0.70, the widest validation-to-held-out gap of any arm. The viable-fitness negatives are a different population from the released split's nonessential genes, and a model that ranks them well does not rank the released genes better. Hypothesis, untested: the released split's nonessential class is enriched for genes with a measurable fitness defect, which the genome-wide negatives are not.

3.5 Gradient norms ✗

On the split-file cells the preprocessor's gradient norm starts near 10^{-2} and rises about thirty-fold over 80 epochs while the head's stays flat near 0.4 (Figure 1d); the clip at 10 never binds. The rise tracks the validation curve's climb, which is the same under-training reading from the gradient side.

4 The metabolic module: three ways to read essentiality off it ✗

The pooled backbone reads its logit from an MLP on the perturbed context (Figure 2). A flux arm builds the enzyme-constrained thermodynamic flux layer on the same context and exposes one physically named quantity, the flux through the growth reaction `r_2111`. Three readouts put that quantity to different uses, and they differ in what the backbone's gradient has to pass through.

biomass (forced) $\ell = a\, v_{\text{growth}} + b$. The growth column is the only path from the context to the logit, so the module carries the whole prediction or the cell scores chance (Figure 3). This is FBA's own reading of essentiality with a learned threshold, and it is the wave-9 design.

context + growth $\ell = \text{MLP}(h) + a\, v_{\text{growth}} + b$. Both paths from the first step; the per-module gradient norms say how much of each update the module carries when it is not forced to carry all of it.

gated $\ell = \text{MLP}(h) + g\,(a\, v_{\text{growth}} + b)$ with $g_0 = 0$ (Figure 4). At initialization the cell is the pooled model exactly; the gate's trajectory, logged every epoch, says whether the optimizer chose the flux path, and the `flux_layer` gradient line says whether gradient reached the layer, which at $g = 0$ it cannot.

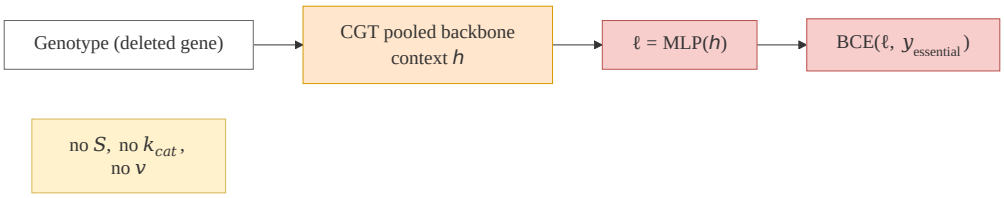


Figure 2: The pooled readout. The backbone alone: one MLP from the perturbed context to a logit, binary cross entropy against the released label. No stoichiometry, no kinetic parameters, no flux. Rendered from `notes/experiments.028-gene-essentiality.mermaid.readouts.md`.

4.1 Round 2: the module on the stable backbone, with and without the penalties ✗

Round 2 (jobs 2182 to 2208) puts the three readouts on the width-12 cell of Section 3, on the released-label split, at three seeds each: the forced and gated readouts at constraint scale 0.1 and 0, the context readout at both scales, the forced readout at scale 0 with the k_{cat} vector permuted across catalytic units (same architecture, no unit-to-value correspondence), and the stoichiometry-only layer (`flux_off`: no thermodynamics, no capacity) with the forced and context readouts. Every flux arm carries the wave-9 flux configuration otherwise: the predicted k_{cat} tier, the union-external thermodynamic source, twenty warmup epochs, the dataset medium, the mixture concentration space. The round's table and figure enter here when its readout runs.

The reads are fixed before the numbers: *not hurt* is the gated arm against the pooled cell paired by seed, with the gate's trajectory beside it; *help* is the forced arm above the pooled cell on three of three seeds and above its shuffled-k_{cat} control; the *collapse's cause* is the forced arm at scale 0.1 against scale 0, since on the betaxanthin targets the penalty scale was the dose that starved the layer's input.

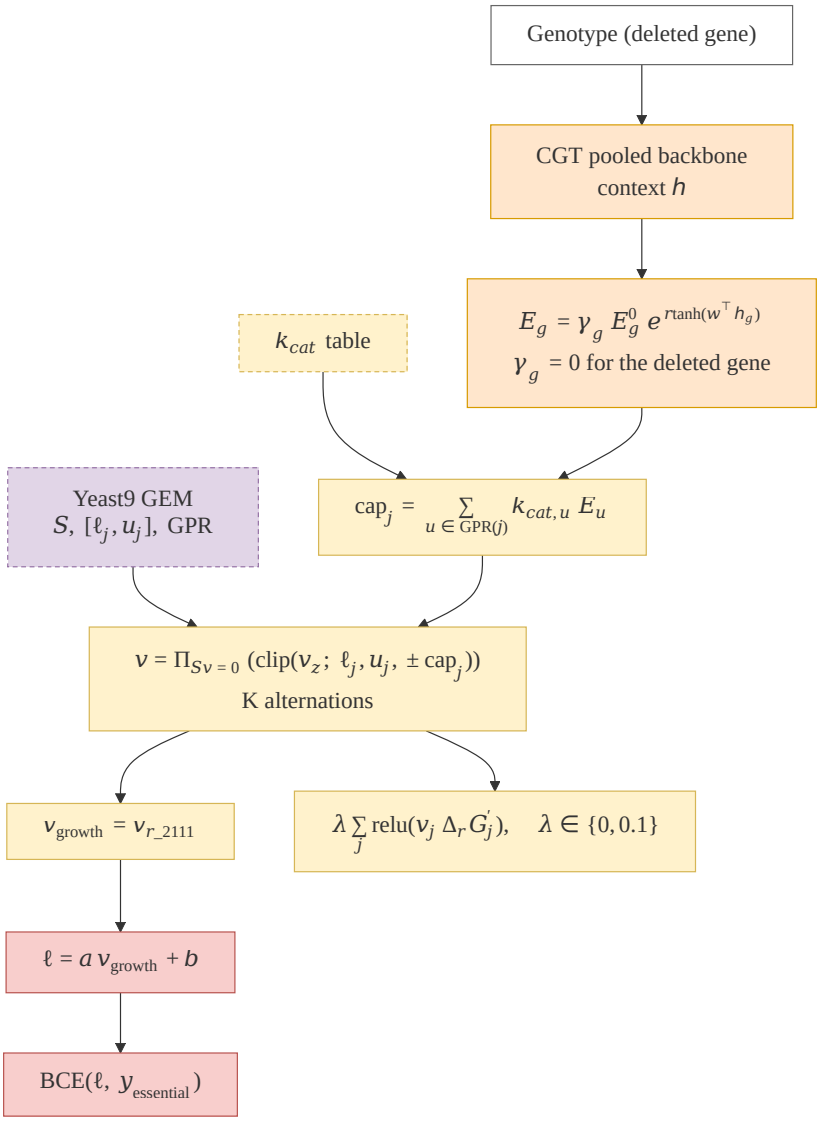


Figure 3: The forced biomass readout. The context sets each enzyme's abundance (a deleted gene's is zero), the k_{cat} table turns abundances into capacities, and the flux is the projection onto $Sv = 0$ of the capacity-clipped proposal; the logit is an affine of the growth column alone. The thermodynamic penalty is a loss on the flux with no path to the prediction, at scale 0.1 or 0.

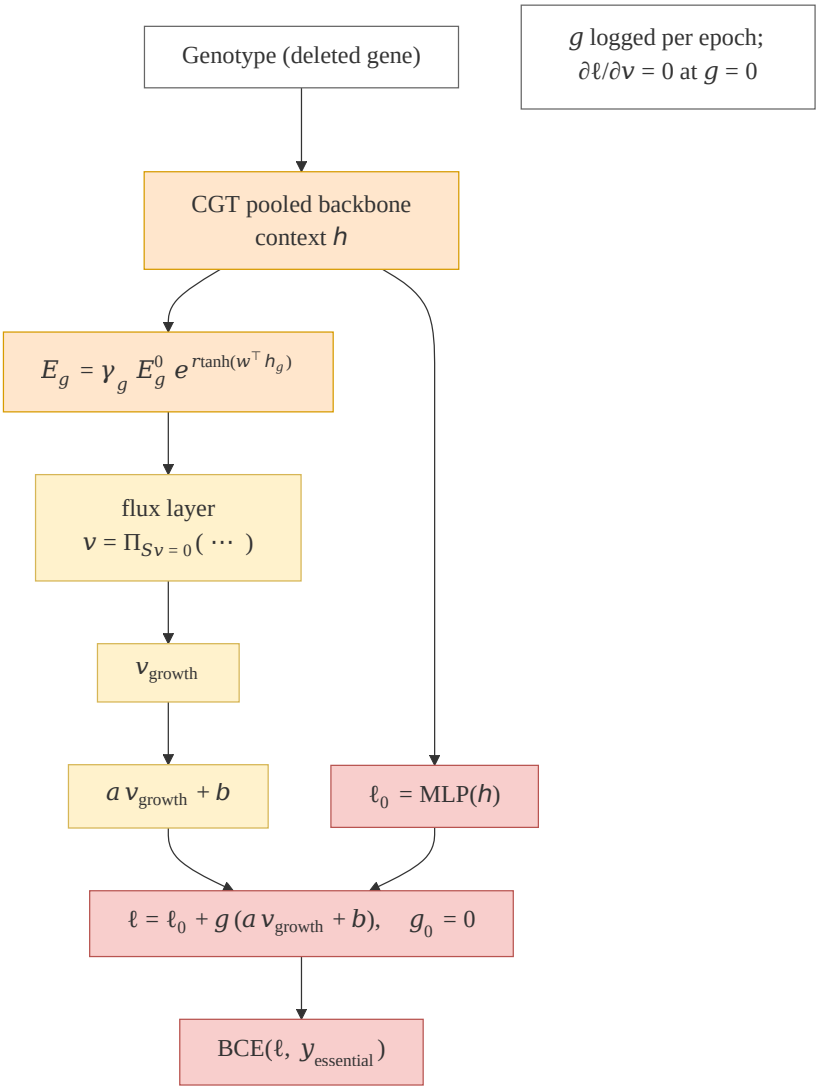


Figure 4: The gated readout. The pooled MLP's logit plus the growth affine behind a gate initialized at zero. The gate is a parameter of the readout; its value is written into every epoch row as `ess_gate`.

5 Open ✗

- Round 3 (jobs 2209 to 2222): batch 32 with the class weight at 80 and 200 epochs, batch 32 alone at 200 epochs, and five fold models on the cross-validation files whose logits are averaged on the held-out genes, which is the protocol behind the published FCL number.
- FBA's growth ratio as an input to the transformer. The ladder's best rung is a logistic regression on a frozen embedding with the four ratios appended (0.803), and neither half reaches that alone; a cell that concatenates the ratios to the readout, or carries them as a node feature, tests whether the transformer gains the same complement.
- The module's own hyperparameters have not been moved on this endpoint: the k_{cat} tier (measured against predicted), the thermodynamic source, the parameterization (projection against nullspace), the number of alternations.
- Every cell here is width 12. The depth-against-width isolation (width 90 at one block, width 12 at two) has not run on the split-file protocol.